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**Unit :3 BAYESIAN AND COMPUTATIONAL LEARNING**

Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model

**Rubrics (Applicable to all Assignments)**

| **Criterion** | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Needs Improvement (2)** | **Poor (1)** |
|---|---|---|---|---|---|
| Problem Understanding | Clearly identifies problem and objectives | Mostly clear | Adequate | Limited understanding | Incorrect understanding |
| Data Collection & Preparation | Appropriate, clean dataset with preprocessing | Minor preprocessing issues | Basic preprocessing | Poor preprocessing | No preprocessing |
| Analysis using NumPy/Pandas | Correct and efficient analysis | Mostly correct | Basic analysis | Limited analysis | Incorrect analysis |
| Visualization & Interpretation | Appropriate graphs with meaningful insights | Good visualization | Basic charts | Minimal interpretation | No visualization |
| Innovation & SDG Mapping | Excellent real-world relevance with SDG linkage | Good linkage | Moderate linkage | Weak linkage | No SDG relevance |
| Documentation & Conclusion | Professional report with justified conclusions | Good report | Adequate report | Poor report | Incomplete |

**Assignment Instructions for Students**

1. Analyze the given real-world problem in the context of the assigned **SDG**.
2. Identify the **Bayesian or computational learning concept** that best addresses the problem.
3. Explain **why the selected algorithm is appropriate**.
4. Describe the **input data, expected output, and evaluation metrics**.
5. Discuss the **societal and sustainability impact** of the proposed solution.

These questions are suitable for **B.Tech/BE Machine Learning** courses, align with **Bloom's Taxonomy (Analyze, Evaluate, Create)**, and connect machine learning concepts with the **United Nations Sustainable Development Goals (SDGs)**.

| No. | Assignment Question | ML Topic | SDG |
|---|---|---|---|
| 1 | A hospital wants to predict diabetes using Bayes' Theorem. Explain how prior and posterior probabilities help doctors make better decisions. | Bayes Theorem | SDG 3 – Good Health & Well-being |
| 2 | Analyze how Naïve Bayes can classify fake and genuine health awareness messages on social media. | Naïve Bayes | SDG 3 |
| 3 | A village health center has limited patient data. Explain why Bayesian Learning is suitable for disease prediction in such situations. | Bayesian Learning | SDG 3 |
| 4 | Design a Bayes-based model to predict student dropout risk using attendance and internal marks. | Bayes Classifier | SDG 4 – Quality Education |
| 5 | Explain how Concept Learning can identify students requiring academic support. | Concept Learning | SDG 4 |
| 6 | Analyze how Maximum Likelihood Estimation can improve online examination performance prediction. | Maximum Likelihood | SDG 4 |
| 7 | Compare Bayes Optimal Classifier and Naïve Bayes for predicting student placement outcomes. | Bayes Optimal | SDG 4 |

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| 8 | Explain how Gibbs Algorithm can improve decision-making in educational recommendation systems. | Gibbs Algorithm | SDG 4 |
| 9 | Develop a Bayesian model to recommend scholarships based on student performance. | Bayesian Belief Network | SDG 4 |
| 10 | Explain how Probability Learning can help personalize online learning platforms. | Probability Learning | SDG 4 |
| 11 | Analyze how Naïve Bayes can classify recyclable and non-recyclable waste using sensor data. | Naïve Bayes | SDG 11 – Sustainable Cities |
| 12 | Design a Bayesian system to predict waste collection requirements in smart cities. | Bayesian Belief Network | SDG 11 |
| 13 | Explain how EM Algorithm can group similar waste generation patterns in urban areas. | EM Algorithm | SDG 11 |
| 14 | Compare Maximum Likelihood and EM Algorithm for predicting urban traffic congestion. | MLE vs EM | SDG 11 |
| 15 | Explain how Sample Complexity affects the accuracy of smart city prediction systems. | Sample Complexity | SDG 11 |
| 16 | Analyze how Bayes' Theorem helps predict flood-prone areas using rainfall history. | Bayes Theorem | SDG 13 – Climate Action |
| 17 | Explain how Bayesian Belief Networks model relationships between rainfall, river levels, and flooding. | Bayesian Belief Network | SDG 13 |
| 18 | Design a Naïve Bayes classifier for predicting air quality levels. | Naïve Bayes | SDG 13 |
| 19 | Explain how Gibbs Algorithm can improve weather prediction under uncertainty. | Gibbs Algorithm | SDG 13 |
| 20 | Analyze how Probability Learning supports climate risk assessment. | Probability Learning | SDG 13 |
| 21 | Explain how Maximum Likelihood can estimate crop disease probability from agricultural data. | Maximum Likelihood | SDG 2 – Zero Hunger |
| 22 | Design a Bayesian model for predicting crop yield based on rainfall and soil quality. | Bayesian Learning | SDG 2 |

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| 23 | Explain how Concept Learning identifies healthy and unhealthy crops. | Concept Learning | SDG 2 |
| 24 | Analyze how EM Algorithm clusters farms with similar irrigation requirements. | EM Algorithm | SDG 2 |
| 25 | Explain how Finite and Infinite Hypothesis Spaces influence crop prediction models. | Hypothesis Spaces | SDG 2 |
| 26 | Design a Bayes classifier to predict machine failures in manufacturing industries. | Bayes Classifier | SDG 9 – Industry & Innovation |
| 27 | Explain how Minimum Description Length Principle prevents overfitting in industrial prediction systems. | MDL Principle | SDG 9 |
| 28 | Analyze the role of Bayesian Belief Networks in predictive maintenance of machines. | Bayesian Belief Network | SDG 9 |
| 29 | Explain how Sample Complexity affects industrial quality prediction models. | Sample Complexity | SDG 9 |
| 30 | Compare Mistake Bound Model and Bayesian Learning for industrial defect detection. | Mistake Bound Model | SDG 9 |
| 31 | Design a Naïve Bayes model for predicting customer loan repayment behavior. | Naïve Bayes | SDG 8 – Decent Work & Economic Growth |
| 32 | Explain how Bayes Optimal Classifier improves financial fraud detection. | Bayes Optimal | SDG 8 |
| 33 | Analyze the role of Bayesian Learning in reducing loan approval bias. | Bayesian Learning | SDG 8 |
| 34 | Explain how Probability Learning predicts customer purchasing behavior in e-commerce. | Probability Learning | SDG 8 |
| 35 | Compare Maximum Likelihood and Bayes' Theorem for credit risk analysis. | Bayes vs MLE | SDG 8 |
| 36 | Explain how Bayesian Learning can improve road accident prediction systems. | Bayesian Learning | SDG 3 & SDG 11 |
| 37 | Design a Bayesian Belief Network for emergency ambulance routing. | Bayesian Belief Network | SDG 3 |
| 38 | Explain how Mistake Bound Models reduce errors in autonomous vehicle decision-making. | Mistake Bound Model | SDG 11 |

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| 39 | Analyze how EM Algorithm improves customer segmentation for sustainable public transport planning. | EM Algorithm | SDG 11 |
| 40 | Propose a complete Bayesian Machine Learning solution for an SDG of your choice. Justify the selected algorithm, expected benefits, required data, and evaluation metrics. | Comprehensive Bayesian Learning | Any SDG |

# ANSWERS

**1. A hospital wants to predict diabetes using Bayes' Theorem. Explain how prior and posterior probabilities help doctors make better decisions.**

Introduction:

Bayes' Theorem is a probability-based method used to predict the possibility of diabetes using patient information such as age, blood sugar level, weight and family history.

Working:

1. Collect patient details such as age, glucose level, BMI and family history.
2. Calculate the prior probability of diabetes based on previous medical records.
3. Observe new patient information and calculate its likelihood.
4. Apply Bayes' Theorem to calculate the posterior probability.
5. Predict whether the patient has a high or low risk of diabetes.

Formula:

$$P(D|E) = \frac{P(E|D) \times P(D)}{P(E)}$$

Where:

- P(D) – Prior probability of diabetes.
- P(E|D) – Likelihood of observing the evidence.
- P(D|E) – Posterior probability of diabetes.
- P(E) – Probability of the evidence.

Importance:

- Helps doctors make better decisions.
- Handles uncertainty in medical data.
- Uses previous medical knowledge.
- Improves early disease prediction.

SDG Integration:
SDG 3 – Good Health and Well-being. It supports early diabetes detection and better healthcare.
Conclusion:
Bayes' Theorem helps doctors predict diabetes by combining previous knowledge with new patient information.

**2. Analyze how Naïve Bayes can classify fake and genuine health awareness messages on social media.**
Introduction:
Naïve Bayes is a classification algorithm that can identify whether health awareness messages shared on social media are fake or genuine.
Working:

1. Collect health-related messages from social media.
2. Label messages as fake or genuine.
3. Preprocess the text by removing unnecessary words and symbols.
4. Extract important words and features.
5. Train the Naïve Bayes classifier using labelled messages.
6. Calculate the probability of each message being fake or genuine.
7. Classify the message based on the highest probability.

Example:

- Genuine message: Information from an authorized health organization.
- Fake message: False medical claims or misleading health information.

Importance:

- Detects misleading health information.
- Helps reduce the spread of fake news.
- Provides fast message classification.
- Supports public health awareness.

SDG Integration:
SDG 3 – Good Health and Well-being. It promotes access to reliable health information.
Conclusion:
Naïve Bayes helps identify fake and genuine health messages and supports the spread of trustworthy health information.

**3. A village health center has limited patient data. Explain why Bayesian Learning is suitable for disease prediction in such situations.**

Introduction:

Bayesian Learning is a machine learning approach that combines prior knowledge and available patient data to predict diseases. It is useful when a village health center has limited medical records.

Working:

1. Collect available patient details and medical history.
2. Use previous medical knowledge as prior probability.
3. Calculate the likelihood of diseases based on symptoms.
4. Apply Bayes' Theorem to update probabilities.
5. Calculate posterior probabilities.
6. Predict the most likely disease.

Importance:

- Works with limited training data when useful prior knowledge is available.
- Handles uncertainty in medical information.
- Combines previous knowledge with new observations.
- Updates predictions when new patient data is collected.
- Supports early disease detection.

Limitations:

- Limited data may reduce prediction accuracy.
- Incorrect prior assumptions can affect results.
- Medical expert guidance is necessary.

SDG Integration:

SDG 3 – Good Health and Well-being. It supports healthcare services in rural and underserved areas.

Conclusion:

Bayesian Learning is suitable for village health centers because it combines available patient data with prior medical knowledge to make disease predictions.

**4. Design a Bayes-based model to predict student dropout risk using attendance and internal marks.**

Introduction:

A Bayes-based classification model can predict student dropout risk by analysing attendance percentage and internal examination marks.

Working:

1. Collect student attendance, internal marks and academic records.
2. Classify students into low-risk, medium-risk and high-risk categories.
3. Calculate prior probabilities for each risk category.
4. Analyse attendance and internal marks.
5. Apply Bayes' Theorem to calculate posterior probabilities.
6. Predict the student's dropout risk.
7. Identify students who require additional academic support.

Example:

| **Student Details** | **Predicted Risk** |
|---|---|
| High attendance, good marks | Low |
| Medium attendance, average marks | Medium |
| Low attendance, poor marks | High |

Importance:

- Identifies students at risk of dropping out.
- Helps teachers provide early support.
- Improves student academic performance.
- Supports educational planning.

SDG Integration:

SDG 4 – Quality Education. It helps reduce student dropout rates and promotes equal learning opportunities.

Conclusion:

The Bayes-based model helps educational institutions identify students at dropout risk and take preventive actions.

**5. Explain how Concept Learning can identify students requiring academic support.**

Introduction:

Concept Learning is a machine learning technique that learns rules from examples. It can identify students who require academic support by analysing their academic performance.

Working:

1. Collect student attendance, internal marks and assignment scores.
2. Prepare labelled examples of students who need support and those who do not.
3. Identify important attributes such as attendance and examination performance.
4. Create a hypothesis space containing possible classification rules.
5. Train the model using positive and negative examples.
6. Identify rules for students who require academic support.
7. Apply the learned rules to new student records.

Example:

| Attendance | Internal Marks | Result |
|---|---|---|
| High | High | No additional support |
| Low | Low | Academic support required |
| Medium | Low | Academic support required |
| High | Medium | Monitor performance |

Importance:

- Identifies academically weak students.
- Helps teachers provide additional coaching.
- Improves student performance.
- Supports early intervention.
- Reduces academic failure.

SDG Integration:

SDG 4 – Quality Education. It promotes inclusive and equal educational opportunities.

Conclusion:

Concept Learning helps identify students who need academic support by learning classification rules from their attendance and academic performance.